TEST REPORT

Mandatory Test Case 1: Normal Start

• Mode: OFF → ACC → IGNITION_ON

• Speed: 40

• Temp: 80

• Gear: 1

Expected: No faults, NORMAL state

Actual:

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 40 80 1 2
[MAIN] Inputs read: speed=40 temp=80 gear=1 mode=2
[MAIN] Inputs validated: speed=40 temp=80 gear=1 mode=2
[MAIN] Mode updated: active=2 previous=1
[MAIN] Control checks done: cycle_flags=00000 priority=0
[MAIN] Fault status updated: persistent=00000
[MAIN] State evaluated: system_state=0

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 40 km/h
  Temperature : 80 C
  Gear        : 1
  Req. Mode   : IGNITION_ON

[MODE]
  Active Mode   : IGNITION_ON
  Current Mode  : IGNITION_ON
  Previous Mode : ACC

[FAULTS]
  Cycle Flags : 00000 [ CLEAR ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 0
  Warnings        : 0
  Critical Faults : 0
  Reset Requested : NO


[STATE]
  System State : NORMAL

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  (none)
```

Observation: PASS

Mandatory Test Case 2: Overspeed

• Speed: 130

• Temp: 85

• Gear: 4

• Mode: IGNITION_ON

Expected: Overspeed detected, fault counter incremented, DEGRADED or warning behaviour as defined

Actual:

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 135 85 4 2
[MAIN] Inputs read: speed=135 temp=85 gear=4 mode=2
[MAIN] Inputs validated: speed=135 temp=85 gear=4 mode=2
[MAIN] Mode updated: active=2 previous=2
[MAIN] Control checks done: cycle_flags=00001 priority=2
[MAIN] Fault status updated: persistent=00000
[STATE] Transition NORMAL -> DEGRADED : Major fault threshold reached (1 >= 1)
[MAIN] State evaluated: system_state=1

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 135 km/h
  Temperature : 85 C
  Gear        : 4
  Req. Mode   : IGNITION_ON

[MODE]
  Active Mode   : IGNITION_ON
  Current Mode  : IGNITION_ON
  Previous Mode : IGNITION_ON

[FAULTS]
  Cycle Flags : 00001 [ OVERSPEED ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 1
  Warnings        : 0
  Critical Faults : 0
  Reset Requested : NO


[STATE]
  System State : DEGRADED

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  3. OVERSPEED
```

Observation: PASS

Mandatory Test Case 3: High Temperature

• Temp: 100

Expected: High temperature warning, no crash, correct priority handling

Actual:

Observation: PASS

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 80 100 3 2
[MAIN] Inputs read: speed=80 temp=100 gear=3 mode=2
[MAIN] Inputs validated: speed=80 temp=100 gear=3 mode=2
[MAIN] Mode updated: active=2 previous=1
[MAIN] Control checks done: cycle_flags=0x0004 priority=1
[MAIN] Fault status updated: persistent=0x0000
[MAIN] State evaluated: system_state=0

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 80 km/h
  Temperature : 100 C
  Gear        : 3
  Req. Mode   : IGNITION_ON

[MODE]
  Active Mode   : IGNITION_ON
  Current Mode  : IGNITION_ON
  Previous Mode : ACC

[FAULTS]
  Cycle Flags : 00100 [ HIGH_TEMP ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 0
  Warnings        : 1
  Critical Faults : 0
  Reset Requested : NO


[STATE]
  System State : NORMAL

[WARNINGS]
  WARNING: HIGH TEMP

[ACTIVE FAULTS - Priority Order]
```

Mandatory Test Case 4: Critical Overheat

• Temp: 115

Expected: Critical fault, state escalation, log reason

Actual:

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 0 115 0 0
[MAIN] Inputs read: speed=0 temp=115 gear=0 mode=0
[MAIN] Inputs validated: speed=0 temp=115 gear=0 mode=0
[MAIN] Mode updated: active=0 previous=1
[MAIN] Control checks done: cycle_flags=00010 priority=4
[MAIN] Fault status updated: persistent=00000
[STATE] Transition NORMAL -> DEGRADED : Major fault threshold reached (1 >= 1)
[MAIN] State evaluated: system_state=1

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 0 km/h
  Temperature : 115 C
  Gear        : 0
  Req. Mode   : OFF

[MODE]
  Active Mode   : OFF
  Current Mode  : OFF
  Previous Mode : ACC

[FAULTS]
  Cycle Flags : 00010 [ CRITICAL_OVERHEAT ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 1
  Warnings        : 0
  Critical Faults : 1
  Reset Requested : NO


[STATE]
  System State : DEGRADED

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  1. CRITICAL_OVERHEAT
```

Observation: PASS

Mandatory Test Case 5: Invalid Gear

• Gear: 9

Expected: Invalid gear fault, safe handling

Actual:

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 0 0 9 0
[MAIN] Inputs read: speed=0 temp=0 gear=9 mode=0
[INPUT] Invalid gear: 9 - retaining last valid (0)
[MAIN] Inputs validated: speed=0 temp=0 gear=0 mode=0
[MAIN] Mode updated: active=0 previous=1
[MAIN] Control checks done: cycle_flags=01000 priority=3
[MAIN] Fault status updated: persistent=00000
[STATE] Transition NORMAL -> DEGRADED : Major fault threshold reached (1 >= 1)
[MAIN] State evaluated: system_state=1

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 0 km/h
  Temperature : 0 C
  Gear        : 0
  Req. Mode   : OFF

[MODE]
  Active Mode   : OFF
  Current Mode  : OFF
  Previous Mode : ACC

[FAULTS]
  Cycle Flags : 01000 [ INVALID_GEAR ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 1
  Warnings        : 0
  Critical Faults : 0
  Reset Requested : NO


[STATE]
  System State : DEGRADED

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  2. INVALID_GEAR
```

Obervation: PASS

Mandatory Test Case 6: Illegal Mode Transition

• OFF → IGNITION_ON directly

Expected: Illegal transition detected, FAULT mode or defined safe behaviour

Actual:

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 80 40 3 2
[MAIN] Inputs read: speed=80 temp=40 gear=3 mode=2
[MAIN] Inputs validated: speed=80 temp=40 gear=3 mode=2
[MODE] Illegal transition 0 -> 2 - forcing FAULT
[MAIN] Mode updated: active=3 previous=0
[MAIN] Control checks done: cycle_flags=0x0000 priority=0
[MAIN] Fault status updated: persistent=0x0000
[STATE] Transition 0 -> 1 : major fault or repeated warnings
[MAIN] State evaluated: system_state=1
 CYCLE SUMMARY REPORT
[INPUT]
  Speed       : 80 km/h
  Temperature : 40 C
  Gear        : 3
  Requested Mode : IGNITION_ON
[MODE]
  Active Mode   : FAULT
  Current Mode  : FAULT
  Previous Mode : OFF
[FAULTS]
  Cycle Flags : 00000 [ CLEAR ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 1
  Warnings        : 0
  Critical Faults : 0
  Reset Requested : NO
[STATE]
  System State : DEGRADED
[ACTIVE FAULTS - Priority Order]
  (none)
```

Observation: PASS

Mandatory Test Case 7: Multiple Faults in One Cycle

• Speed: 140

• Temp: 120

• Gear: 9

Expected: Multiple faults set, priority respected, state escalated

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 140 120 9 2
[MAIN] Inputs read: speed=140 temp=120 gear=9 mode=2
[INPUT] Invalid gear: 9 - retaining last valid (0)
[MAIN] Inputs validated: speed=140 temp=120 gear=0 mode=2
[MAIN] Mode updated: active=2 previous=1
[MAIN] Control checks done: cycle_flags=01011 priority=4
[MAIN] Fault status updated: persistent=00000
[STATE] Transition NORMAL -> DEGRADED : Major fault threshold reached (2 >= 1)
[MAIN] State evaluated: system_state=1

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 140 km/h
  Temperature : 120 C
  Gear        : 0
  Req. Mode   : IGNITION_ON

[MODE]
  Active Mode   : IGNITION_ON
  Current Mode  : IGNITION_ON
  Previous Mode : ACC

[FAULTS]
  Cycle Flags : 01011 [ OVERSPEED CRITICAL_OVERHEAT INVALID_GEAR ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 2
  Warnings        : 0
  Critical Faults : 1
  Reset Requested : NO


[STATE]
  System State : DEGRADED

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  1. CRITICAL_OVERHEAT
  2. INVALID_GEAR
  3. OVERSPEED
```

Observation: PASS

Mandatory Test Case 8: Persistent Fault Across Cycles

• same critical fault repeated for 3 cycles

Expected: fault count increases, persistent behaviour visible, SAFE state reached if rule

Requires

Actual:

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 80 140 3 2
[MAIN] Inputs read: speed=80 temp=140 gear=3 mode=2
[MAIN] Inputs validated: speed=80 temp=140 gear=3 mode=2
[MAIN] Mode updated: active=2 previous=1
[MAIN] Control checks done: cycle_flags=00010 priority=4
[MAIN] Fault status updated: persistent=00000
[STATE] Transition NORMAL -> DEGRADED : Major fault threshold reached (1 >= 1)
[MAIN] State evaluated: system_state=1

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 80 km/h
  Temperature : 140 C
  Gear        : 3
  Req. Mode   : IGNITION_ON

[MODE]
  Active Mode   : IGNITION_ON
  Current Mode  : IGNITION_ON
  Previous Mode : ACC

[FAULTS]
  Cycle Flags : 00010 [ CRITICAL_OVERHEAT ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 1
  Warnings        : 0
  Critical Faults : 1
  Reset Requested : NO


[STATE]
  System State : DEGRADED

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  1. CRITICAL_OVERHEAT
```

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 80 140 3 2
[MAIN] Inputs read: speed=80 temp=140 gear=3 mode=2
[MAIN] Inputs validated: speed=80 temp=140 gear=3 mode=2
[MAIN] Mode updated: active=2 previous=2
[MAIN] Control checks done: cycle_flags=00010 priority=4
[MAIN] Fault status updated: persistent=00000
[STATE] Transition DEGRADED -> SAFE : Critical fault count escalated from DEGRADED (2 >= 2)
[MAIN] State evaluated: system_state=2

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 80 km/h
  Temperature : 140 C
  Gear        : 3
  Req. Mode   : IGNITION_ON

[MODE]
  Active Mode   : IGNITION_ON
  Current Mode  : IGNITION_ON
  Previous Mode : IGNITION_ON

[FAULTS]
  Cycle Flags : 00010 [ CRITICAL_OVERHEAT ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 2
  Warnings        : 0
  Critical Faults : 2
  Reset Requested : NO


[STATE]
  System State : SAFE

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  1. CRITICAL_OVERHEAT
```

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 80 140 3 2
[MAIN] Inputs read: speed=80 temp=140 gear=3 mode=2
[MAIN] Inputs validated: speed=80 temp=140 gear=3 mode=2
[MAIN] Mode updated: active=2 previous=2
[MAIN] Control checks done: cycle_flags=00010 priority=4
[MAIN] Fault status updated: persistent=00010
[MAIN] State evaluated: system_state=2

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 80 km/h
  Temperature : 140 C
  Gear        : 3
  Req. Mode   : IGNITION_ON

[MODE]
  Active Mode   : IGNITION_ON
  Current Mode  : IGNITION_ON
  Previous Mode : IGNITION_ON

[FAULTS]
  Cycle Flags : 00010 [ CRITICAL_OVERHEAT ]
  Persistent  : 00010 [ CRITICAL_OVERHEAT ]
  Major Faults    : 3
  Warnings        : 0
  Critical Faults : 3
  Reset Requested : NO


[STATE]
  System State : SAFE

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  1. CRITICAL_OVERHEAT
```

Observation: PASS

Mandatory Test Case 9: Recovery / Reset Logic

• return values to normal after prior warnings

Expected: team-defined recovery behaviour documented clearly

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 80 140 3 2
[MAIN] Inputs read: speed=80 temp=140 gear=3 mode=2
[MAIN] Inputs validated: speed=80 temp=140 gear=3 mode=2
[MAIN] Mode updated: active=2 previous=2
[MAIN] Control checks done: cycle_flags=00010 priority=4
[MAIN] Fault status updated: persistent=00010
[MAIN] State evaluated: system_state=2

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 80 km/h
  Temperature : 140 C
  Gear        : 3
  Req. Mode   : IGNITION_ON

[MODE]
  Active Mode   : IGNITION_ON
  Current Mode  : IGNITION_ON
  Previous Mode : IGNITION_ON

[FAULTS]
  Cycle Flags : 00010 [ CRITICAL_OVERHEAT ]
  Persistent  : 00010 [ CRITICAL_OVERHEAT ]
  Major Faults    : 3
  Warnings        : 0
  Critical Faults : 3
  Reset Requested : NO


[STATE]
  System State : SAFE

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  1. CRITICAL_OVERHEAT
```

```
--- NEW CYCLE ---

[Input] Enter Speed, Temp, Gear, Mode (0=OFF, 1=ACC, 2=IGN_ON, 3=FAULT): 0 0 0 0
[MAIN] Inputs read: speed=0 temp=0 gear=0 mode=0
[WATCHDOG] 2 continuous SAFE cycles detected. Triggering system reset...
[STATE] System initialised: state=NORMAL, mode=OFF
[MAIN] Inputs validated: speed=0 temp=0 gear=0 mode=0
[MAIN] Mode updated: active=0 previous=0
[MAIN] Control checks done: cycle_flags=00000 priority=0
[MAIN] Fault status updated: persistent=00000
[MAIN] State evaluated: system_state=0

---- CYCLE SUMMARY REPORT ---

[INPUT]
  Speed       : 0 km/h
  Temperature : 0 C
  Gear        : 0
  Req. Mode   : OFF

[MODE]
  Active Mode   : OFF
  Current Mode  : OFF
  Previous Mode : OFF

[FAULTS]
  Cycle Flags : 00000 [ CLEAR ]
  Persistent  : 00000 [ CLEAR ]
  Major Faults    : 0
  Warnings        : 0
  Critical Faults : 0
  Reset Requested : YES


[STATE]
  System State : NORMAL

[WARNINGS]

[ACTIVE FAULTS - Priority Order]
  (none)
```

Observation: PASS